

MATH5320 – Homework II

1)

Community banks are *smaller, and more locally oriented depository institutions that focus on relationship banking with local deposits funding local loans* that are generally diversified, but, especially *SMEs and real estate*. FDIC's "community bank" categorisation" characterises these as "relationship banking, local footprint, traditional intermediation", with the business model being primarily via maturity transformation involving shorter term deposits funding longer-term loans. Under FDIC Quarterly Banking Profile reporting, as of Q3 2025, have identified 3,953 community banks with up to \$2.7 T in assets across the States. Note that the risks that entail community banks should be similar to other commercial and retail banks, but *face more concentration market risk*, as for example, with SVB bank which primarily offered services to start-ups, or CRE (commercial real-estate) concentration at regionals.

Savings & Loans, or thrifts, *operates under savings association charters, historically specialised in residential mortgage lending funded by deposits*, and are structured under federal savings associations. In many ways, it is like a credit union, but it is *mutually held*, but has far more mortgage-heavy balance sheet, meaning it has a high exposure to interest-rate risk and prepayment risk, where borrowers will refinance when rates fall. As a result, in 1965 with ~6000 S&Ls & \$129B assets has led to 2021 with 608 institutions ~1.5T assets, and recently, 2024 with 551 institutions ~\$1.2 in Assets, indicating substantive consolidation in this field, but also a general decline in assets held by S&Ls after crises like the Lincoln Savings & Loans scandal, that was emblematic of the 1980s – 90s after Volcker raised overnight rates to 20%. Note that these statistics are derived from Lecture slides, and the FDIC QBP book.

Credit-unions are *member-owned, non-for-profit financial cooperatives that provides retail banking services* for its members like teachers, plumbers, etc., where members are effectively "owners" under one-membership & one-vote. Net income is often return through better deposit or loan rates and lower fees. Credit Unions have as an industry, stagnated with ~5,000 CUs, \$2.1T assets, 131M members in 2021 to ~4,600 CUs, \$2.3T assets, 143 M members in 2024. They face similar balance sheet risks, like credit, liquidity and interest rate, but governance and customer base makes them operationally different from banks.

Overall, *all retail banks suffer from potential risks* like loan losses that encapsulate issues like risk of pre-payment, default rates and concentration risks (especially for Community banks), demand for deposits (which represent the *supply* of the banks funds), maturity transformation, which requires a balance of keeping enough liquid cash to fulfill withdrawal requirements, but also investing in loans that provide a higher return. However, *all retail banks are FDIC-insured* for a deposit limit of \$250,000 USD per

account, whereas credit unions are insured by the NCUA. All banks, undertake some interest rate risk, as their holdings are fundamentally affected by the interest rates, as, under Dodd-Frank risk-retention rules, it ensured that banks apart from origination, now have to take on some risk themselves, rather than repackage all loans out as ABS, to prevent moral hazard of originating subpar loans. Fundamentally, the clean differentiators are ownership and governance structures like shareholder-owned banks vs member-owned, typical asset mixes (like thrifts being more mortgage-heavy), and different regulatory perimeters (FDIC deposit insurance for banks, NCUA for credit unions).

2)

Investment banks, *primarily service institutional clients, like pension funds, companies, etc., and are more focused on creating financial instruments and analysing financial deals*, rather than deposit-funded loan origination. The functions of Investment banks that are not part of traditional commercial banking (deposit-taking + loan origination) capabilities. They include, but are not limited to, Underwriting & Distribution (ECM/DCM), M&A advisory and restructuring and Market-Making under the Global Markets division for most banks, or call FICC at Goldman (usually under the Investment Bank).

ECM/DCM, is the issuance (organising equity & debt offerings of clients), organised by investment banks, that also include the building of syndicates, marketing the securities, price deals, and ultimately distribute to earn fees. In the US, public offerings involving a FINRA member are subject to FINRA's Corporate Financing review process / filings under Public Offering Rule 5110, or specifically, the practical "file for review / no-objections" pathway. In here, notable risks are market or underwriting risk, namely the IBD has to hold inventory in adverse moves, liquidity risk of being "stuck" with allocations in weaker markets, and also operational and potential regulatory risks with disclosure processes and conflicts.

M&A advisory, refers to IBDs advising buyers/sellers on mergers, acquisitions, divestitures, LBOs, and restructuring. Specifically, IBDs provide valuation, process management, negotiation support, and capital structure designs, and earn advisory fees as a % rather than interest spread. Notably, Goldmans Sachs usually leads the M&A tables, and had still done so by 2025 by deal value, as cited from LSEG. There is less risk balance sheet risk than trading, but there remain meaningful reputational, and regulatory risks, with mandates contingent on trust and conflict management.

Market making, refers to the provision of liquidities (providing bid/ask for products) and hedging capacities across equities, rates, credit and derivatives. Their core risks skew toward market risk such as not hedging delta frequently enough and hedging slippage,

counterparty risk with derivative exposures, and funding/liquidity risk (rather than predominantly credit-on-loans). Investment Banks have been relegated to this mostly Market-making within Global Markets (and others like Prime Brokerage), since the Volcker Rule post-GRC that restricted taking on risk, although some banks like Macquarie Capital still does prop trading in commodities.

Overall, Commercial banks primarily manage credit-on-loans and deposit liquidity via spread income (basically rate they lend – IR they give to depositors), investment banks are built around market structure roles like origination, distribution and liquidity provision, with naturally dominant risks drifting towards market, funding, and counterparty risks.

3)

Broker Dealers	Goldman Sachs (Source) Morgan Stanely (Source) Interactive Brokers (Source)
Exchanges	New York Stock Exchange (Source) NASDAQ (Source) Chicago Mercantile Exchange (Source)
Clearing Houses	National Securities Clearing Corporation (NSCC) Options Clearing Corporation (OCC) Fixed Income Clearing Corporation (FICC) Source: SEC Registered Cleaning Agencies

4)

Regulators

US regulatory agencies include:

- Federal Deposit Insurance Corporation (FDIC) and The National Credit Union Administration (NCUA)
 - US Treasury — Office of Comptroller of the Currency (OCC)
 - Federal Reserve Banks (FRB)
 - Securities and Exchanges Commission (SEC)
 - Commodity Futures Trading Commission (CFTC)
 - Federal Housing Finance Agency (FHFA)
 - US Treasury department's Office of Financial Research (OFR)
 - Consumer Financial Protection Bureau (CFPB)
 - Financial Stability Oversight Council (FSOC) — from Dodd-Frank
 - Financial Industry Regulatory Authority (FINRA)
 - State banking regulators <http://www.csbs.org/>
 - State insurance commissioners <http://naic.org/>
- Financial Crimes Enforcement Network (FinCen) for AML/BSA
 - Office of Foreign assets Control (OFAC) for sanctions control

These latter two are included for completeness, as a *bonus material*, since sanctions have increasingly become more prominent across the past years, most notably since the Russian-Ukrainian War, with sanctions of Russians across the West.

Referring to the above list, and using the abbreviations, we have the following:

Regional Bank	Notably, for regional banks, it is charter dependent. • OCC, if it is a national bank; Fed if it is a state-member bank; FDIC if it is a state non-member bank • State bank regulators, as applicable, if state-chartered • Federal Deposit Insurance Corporation (FDIC) for deposit insurance • CFPB if bank assets exceed \$10B, if it covers consumer financial products • FSOC/OFR monitors for systemic risks, and may coordinate if such risks arise; • Extra: FinCEN & OFAC for BSA & AML
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National Bank	• OCC as the primary prudential supervisor • FDIC for insurance • Federal Reserve (FRB) for supervision of BHCs, but also reserve/payments system role • CFPB for consumer-compliance supervision • FSOC/OFR may become relevant, if systemic risks arise • Extra: FinCen/OFAC for Sanctions/AML
Bank Holding Company	• FRB (consolidated supervision of BHCs) • OCC/FDIC/state regulators at the subsidiary level, depending on charter • CFPB for consumer-compliance • FSOC/OFR may become relevant, but usually will monitor for systemic risks after passing of Dodd-Frank • Extra; FinCen/OFAC for Sanctions/AML
US branches of foreign banks	• Federal Reserve (FRB) (supervises foreign banking orgs in the US) • OCC if federal branches/agencies; state banking regulators if state-licensed • FDIC <i>if</i> they take insured deposits • CFPB if large enough + consumer-facing • Extra: FinCen/OFAC for Sanctions/AML
Hedge fund	• SEC regulates the fund via advisor or the securities part • CFTC is the manager is a CPO/CTA (deal with commodities interests + swaps) ~ depends what type of “fund” • FSOC/OFR monitors for systemic risks, and may step-in to prevent LTCM-like events (unlikely, but in syllabus, and technically ha scope)

	• Extra: FinCen/OFAC for Sanctions/AML (but depends on the specific nature of HF & if it is a master feeder structure)
Broker/Dealer	• SEC is the primary regulator • FINRA is the SRO layer under SEC that regulates broker-dealers • CFTC applies if the firm is active as a derivatives intermediary • Extra: Others like Municipal Securities Rulemaking Board (MSRB) if it does Muni business, NFA if dual-registered for derivatives

5)

Despite both pension plans and insurance companies investing current funds for long-term future payouts, so both face asset-liability mismatch and long-horizon uncertainty, but pension plans tend to have more financial troubles than insurance companies. This is fundamentally because insurers are run inside a much tighter capital/solvency framework, whereas pension funds often have far weaker funding discipline and stronger incentives to underfund.

Insurers have *far stricter solvency discipline with capital triggers and legal-mandated takeover powers*. State insurance regulation has *thresholds like risk-based capital that are tied to the riskiness of assets/operations and gives regulators intervention powers if capital becomes severely “weak”*, and as such, there is far more corrective action before insolvency become unavoidable, as pension plans have no such regulations. NAIC provides more information [here](#). This links to the regulatory risks that entail pension plans, or lack thereof. For instance, consider the General Motors Pension Plan, when there was a potential for full pension insolvency, only mitigated through a government-backed restructuring, that still resulted in [benefit cuts of up to 70%](#) for some retirees. Without such an intervention from the Federal government, it is likely the GM Pension Plan would have been fully insolvent, and notably, if subject the same solvency discipline as Insurers, would not have ended in the situation.

Pension plans are also chronically underfunded, because the sponsor like city or corporation, directly control contributions, and there is a fundamental moral hazard of “kicking the can down the road”. Sponsors have an incentive to delay contributions, especially when budgets are tight, and management wants to preserve cash, where unrealistic promises are made for the future in negotiations. For instance, Detroit’s bankruptcy plan still involved pension cuts and elimination or reduction of cost-of-living adjustments, indicating that when the primary sponsor fails, the “promise” becomes no-longer set in stone. As such, the sponsor-linked funding is a direct moral hazard channel that threatens pension plans, but not insurance companies.

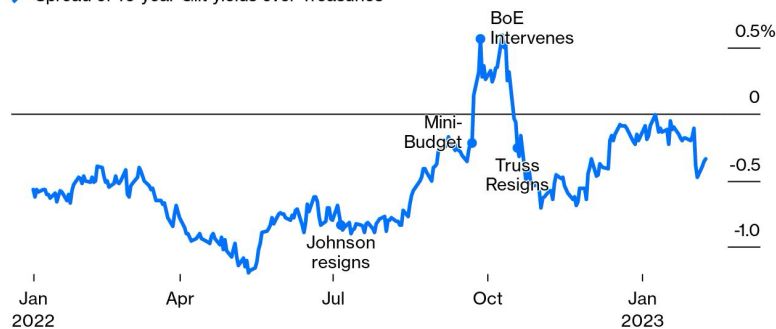
Furthermore, Pension liabilities are “sticky” and hard to reprice, whereas insurers can reprice new business on an annual basis for MOST policies. Insurers often adjust premiums, underwrite standards, and their product mix over time, especially for new businesses, whereas DB pension plans literally cannot “reprice” the accrued benefits without resistance. Such benefits promises are contractual and often political, and incredibly hard to restructure, unless during times of distress.

Finally, liquidity stress can hit both insurers and pension plans quite suddenly, despite long-term hedging. But pension plans show up more, because many DB pension plans use leverage LDI to hedge long-rate or inflation exposures and have slower operational and governance mechanisms, whereas insurers although still exposed, have together liquidity supervision (hence not forced to sell at bad times), and can reprice over time.

The Gilts Meltdown: A Timeline

UK bonds suddenly parted company with the US after the mini-budget

— Spread of 10-year Gilt yields over Treasuries



Source: Bloomberg

BloombergOpinion

A notable example was under Liz Truss in the UK with the “mini budget” that blew up gilt yields in Sept-22, LDI funds faced large collateral calls, when gilts were sold, and pushed prices further down that threatened the viability of British pension plans, as gilt markets became illiquid. The Bank of England’s Financial Stability Report explicitly describes the spiral and estimates collateral calls more than £70bn, explicitly noting that operational lags and pension-scheme governance processes hindered timely

replenishment of liquidity buffers, and as such, the BoE had to backstop for the pension plans. Insurers can face similar liquidity stress, like AIG in 2008, when value of referenced CDOs fell, and AIG had to post increasing collateral on CDS; it drained liquidity and forced intervention. The key contrast is that insurers tend to have stricter liquidity supervision, and hence, are not “forced to sell low in illiquid markets”, whereas pension plans are less operationally agile from funding frictions and governance delays.

Overall, pension plans run into trouble more often, *as their funding is sponsor-controlled and thus there is a moral hazard of underfunding, liabilities are rigid and difficult to reprice, whereas some common hedging approaches and operational rigidities can generate sudden stresses.* Conversely, *insurers face similar economic risks, but regulatory environment imposes stricter solvency discipline, and the failure management system of receivership and guarantees, is defined to preserve continuity.*